

Wikipedia & ZIM Statistics Analysis

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Wikipedia Size Statistics

English Wikipedia (As of April 2025)

- **Articles:** ~6.8 million
- **Total Pages:** ~56.5 million (including talk pages, user pages, etc.)
- **Database Size:** ~190 GB (text only, uncompressed)
- **Media Size:** ~7.5 TB (images, audio, video)
- **ZIM File Size:**
 - Text-only: ~93 GB compressed
 - With images (selection): ~120-160 GB
 - Full media: Not typically distributed as single ZIM due to size

Interlinking Statistics

- **Average Links per Article:** ~92 internal links
- **Total Internal Links:** ~625 million
- **Category Relationships:** ~1.8 million categories with hierarchical structure
- **Template Usage:** ~45,000 templates used across articles

Knowledge Graph Considerations

Neo4j Scaling Limitations

- Single-instance Neo4j practically limited to ~100 billion relationships
- Full Wikipedia graph would contain:
 - ~6.8 million nodes (articles)
 - ~625 million relationships (just from direct links)
 - Additional relationships from categories, templates, infoboxes, etc.

Distributed Knowledge Graph Approaches

Sharding Options

1. **Topic-based Sharding:** Partition by major categories
 - Pro: Natural content boundaries
 - Con: Uneven shard sizes, cross-domain topics problematic
2. **Alphabetical/Hash-based Sharding:** Distribute evenly by article title or ID
 - Pro: Even distribution
 - Con: Related content often separated across shards
3. **Graph Partitioning Algorithms:** Use community detection to minimize cross-shard links

- Pro: Preserves semantic relationships
- Con: Computationally expensive to create and maintain

Practical Implementation Challenges

- Cross-shard query performance
- Consistency across distributed instances
- Synchronization of updates
- Resource requirements (memory-intensive)

User Browsing Patterns

- **Average Session Length:** 3-4 articles per session
- **Navigation Depth:** 75% of users go no deeper than 3 links from entry
- **Topic Clustering:** Users typically stay within related topic clusters
- **Popular Entry Points:** ~5% of articles account for ~40% of all entry points

Recommendations for Distributed Approach

Based on browsing patterns, consider:

1. **Localized Graph Segments:** Create subgraphs around popular entry points
2. **Dynamic Loading:** Load subgraphs based on user navigation paths
3. **Pre-computed Relationships:** Store high-value relationships (similarity, conceptual links)
4. **Hybrid Storage:** Use graph DB for current session context, traditional DB for broader content
5. **Edge Computing:** Consider partial graph processing on client devices

This approach would preserve the benefit of knowledge graph relationships while being practically implementable with current technology constraints.

ZIM Reader Statistics

Performance Benchmarks

- **Startup Time:** ~0.8-2.5 seconds
- **Article Load Time:** ~0.2-0.4 seconds
- **Search Index Generation:** ~3-8 hours for full English Wikipedia
- **Memory Usage:** ~120-250 MB baseline, ~350-500 MB during active browsing

User Experience Metrics

- **Responsiveness Threshold:** >1 second article load perceived as slow
- **Search Response Expectation:** <0.5 seconds for suggestions, <2 seconds for results